Exam-Slot Assignment with QAOA and Simulated Annealing: Part II

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Contents

1 Strategy

2 EMF chain

3 BESSER chain

4 Validation

5 Limitations

A Rule-to-reference traceability

B Full results

C Metamodels

D Test instances

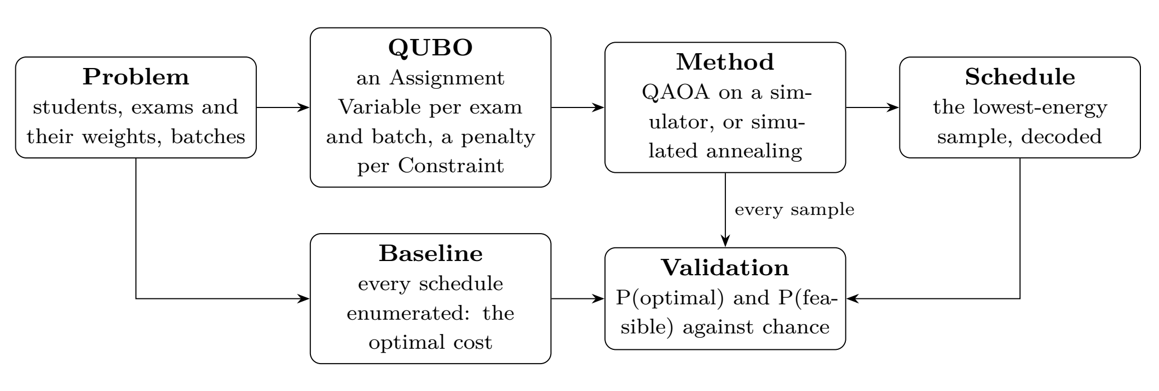
E Reproducing the results

1 Strategy

Terms are those of the EMF chain's glossary, `emf/GLOSSARY.md`.[1]

Missing: The strategy, independent of either chain. Waits on the team.

- The problem: exams into a fixed number of batches so that no student sits two exams of the same batch; an exam's weight is its priority.
- The pipeline of Fig. 1, shared by both chains.
- The two formulations: the paper's [7, §5.3, Eqs. (5)–(10)] (each exam in exactly one batch, penalties only), which BESSER implements, and the weighted one (each exam in at most one batch, weights as priorities; `emf/docs/adr/0009`), which EMF implements. On four mutually conflicting exams in three batches (BESSER's `none`, EMF's `clique`) their optima differ: one conflict accepted, or Statistics left unscheduled at cost 1.
- Why simulated annealing as well: the paper samples by quantum annealing on D-Wave hardware; the chains use its classical counterpart, as the notebook [5, cell 15] does (quantum annealing itself is issue #23).



Problem
students, exams and their weights, batches
QUBO
an Assignment Variable per exam and batch, a penalty per Constraint
Method
QAOA on a simulator, or simulated annealing
Schedule
the lowest-energy sample, decoded
every sample
Baseline
every schedule enumerated: the optimal cost
Validation
P(optimal) and P(feasible) against chance

Figure 1: The strategy of both chains: model the Problem, formulate it as a QUBO, sample it by a Method, decode the lowest-energy sample into a Schedule, and judge every sample against a Baseline that enumerates every schedule.

2 EMF chain

Missing: The EMF chain. Waits on the team.

- Xtext: one `.exam` file holds a Problem and its Solver Configuration (Section D); every model is checked against its metamodel's OCL (Section C) between steps.
- ATL (EMFTVM), Problem → Conflict Graph: an edge between two exams sharing a student [7, §5.2, Algorithm 1].
- Acceleo, Conflict Graph and Solver Configuration → `solve_<name>.py`: the data, then the notebook's pyqubo formulation [5, 6] made a weighted set packing [2, §4.2].
- The runtime samples the compiled model by simulated annealing or by QAOA [1] on Qiskit Aer [3], decodes the lowest-energy sample into a Schedule model and runs the Baseline.
- Every rule, template section and runtime step cites its reference next to it, in the code; Table 2 lists them.

3 BESSER chain

Missing: The BESSER chain. Waits on the BESSER rework.
T1 (Problem → QUBO, rules R1–R6), T2 (QUBO → QAOA circuit in BESSER's quantum metamodel, rules Q1–Q7), T3 (BESSER's `QiskitGenerator`) and the mapping of samples back to Schedules, with the reference of each rule. BESSER's code cites the paper as `[Zhou]`; with `[Paper …]` tags, Section A could list its rules too.

4 Validation

Missing: How the chains are validated. Waits on the team.
What every run measures: the Schedule found, whether it is feasible and optimal against the Baseline, P(optimal) and P(feasible) over every shot or read, and the odds of a uniformly random bit string, the bar a Method must clear. The EMF chain also checks its formulation in every run: a sample's energy is minus the weight it places exactly when it breaks no Constraint. Each chain is measured on its own instances (Section D); every run is in Section B.

Missing: These EMF numbers come from a local run, not from CI. Waits on CI.
Once CI has uploaded an `emf-results` artifact (push #24 and #25), rerun `build.sh`, or `build.sh --run <id>` for a pull request's run.

Table 1: P(optimal) of the EMF chain on each test instance (Section D) by Method, against a uniformly random bit string's; in brackets, how many times chance. Solver Configurations in Table 3; from a local run.

Instance	Variables	Random	QAOA	Simulated annealing
path	8	0.0078	0.106 (14×)	0.940 (120×)
cycle	15	0.00092	0.033 (36×)	1.000 (1092×)
clique	12	0.0015	0.042 (28×)	0.940 (642×)
semester	18	0.000046	0.00024 (5×)	0.960 (20972×)

Missing: The BESSER chain's table, P(optimal) against chance on its instances. Waits on the BESSER rework.
From `besser/output/<instance>/6_validation.json` once the chain settles.

5 Limitations

Missing: The limitations. Waits on the team.

- QAOA is far behind simulated annealing (Table 1), and only simulated, on at most 18 qubits.
- The QAOA runtime starts from a fixed ramp of angles, which follows the annealing path only when γ and β have opposite signs; flipping the mixer's sign raised P(feasible) everywhere and P(optimal) in 8 of 9 cases (`docs/assets/grilling-report-2.md`, §6; not yet fixed).
- The simulated annealing is classical: no quantum annealer was used (issue #23).
- The Baseline enumerates every schedule, so validation stops at about 10^6 schedules.
- The Schedule the generated script writes is not validated against its metamodel's OCL.
- The BESSER chain's own.

References

[1] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. *A Quantum Approximate Optimization Algorithm.* 2014. arXiv: `1411.4028v1` (cited in § 2, § A).

[2] Fred Glover, Gary Kochenberger, and Yu Du. *Quantum Bridge Analytics I: A Tutorial on Formulating and Using QUBO Models.* Published in 4OR 17 (2019) 335–371. 2019. arXiv: `1811.11538v6` (cited in § 2, § A).

[3] Ali Javadi-Abhari et al. *Quantum computing with Qiskit.* 2024. arXiv: `2405.08810` (cited in § 2, § E).

[4] Andrew Lucas. "Ising formulations of many NP problems". In: *Frontiers in Physics* 2 (2014), p. 5. DOI: `10.3389/fphy.2014.00005`. arXiv: `1302.5843v3` (cited in § A).

[5] *`QuantumExamScheduling.ipynb`*. The reference notebook of the paper's formulation, in the repository as `emf/QuantumExamScheduling.ipynb`; cells are numbered by their execution counts (cited in § 1, § 2, § A).

[6] Mashiyat Zaman, Kotaro Tanahashi, and Shu Tanaka. "PyQUBO: Python Library for Mapping Combinatorial Optimization Problems to QUBO Form". In: *IEEE Transactions on Computers* 71.4 (2022), pp. 838–850. DOI: `10.1109/TC.2021.3063618` (cited in § 2, § E).

[7] Ziyu Zhou et al. "Hybrid quantum annealing for large-scale exam scheduling: Validation in real-world educational scenarios". In: *Applied Soft Computing* 184 (2025), p. 113756. DOI: `10.1016/j.asoc.2025.113756` (cited in § 1, § 2, § A).

A Rule-to-reference traceability

Table 2: Every ATL rule, template section and runtime step of the EMF chain that implements the formulation or QAOA, with the reference its comment cites, read from the code.

Unit	What it implements, as its comment says
ATL, `problem2conflictgraph.atl`: Problem → Conflict Graph	
`rule Problem2ConflictGraph`	[7, §5.2, Fig. 3]: the conflict graph G, exams as nodes and conflicts as edges.
`rule Problem2ConflictGraph`	[7, §5.3, Eq. (4)]: X has only M columns, so the M batches bound the schedule.
`rule Exam2Exam`	[7, §5.2, Fig. 3]: each exam is a node of G.
`rule Exam2Exam`	[2, §4.2]: the weight w_j of the set packing objective, the exam's priority (ADR-0009).
`rule NewBatch`	[7, §5.3, Eq. (4)]: batch k is column k of X, k = 1..M.
`helper sharesStudentWith`	[7, §5.2, Algorithm 1, Eq. (3)]: C[i,j] = 1 iff some student sits both exam i and exam j.
`lazy rule Conflict`	[7, §5.2, Fig. 3]: one edge of G per pair with C[i,j] = 1, emitted once, in problem order.
Acceleo, `main.mtl`: Conflict Graph and Solver Configuration → `solve_<name>.py`	
`CONFLICTS`	[7, §5.2, Algorithm 1, Eq. (3)] [5, cell 7]: C, the contradiction matrix, as pairs (i, j), i < j.
`x`	[7, §5.3, Eq. (4)] [5, cell 11]: x[i, k] = 1 iff exam i is in batch k + 1
`constraint1`	[2, §4, known penalties, row 5] [4, §4.2, Eq. (30)]: each exam in at most one batch, the paper's exactly-one [7, §5.3, Eqs. (5)–(7)] relaxed (ADR-0009). With one batch there is no pair to forbid.
`constraint2`	[7, §5.3, Eqs. (8)–(9)] [5, cell 12]: conflicting exams never share a batch, the penalty x[i, k] x[j, k] of [2, §4, known penalties, row 1].
`profit`	[2, §4.2] [4, §4.2, Eq. (31)]: scheduling exam i earns its weight; an unscheduled exam costs it
`A, H, model`	[7, §5.3, Eq. (10)] [5, cell 13]: H, with lambda = gamma = A and the profit added; A is fed from the Solver Configuration
Python, `examschedule.py`: the runtime the generated scripts call	
`schedule_of`	[7, §5.5]: the batch of each exam read off X, x[i, k] = 1 iff exam i is in batch k + 1.
`penalty`	A: the largest weight plus one unless set. An explicit A must exceed every weight, or breaking a constraint can cost less than leaving an exam unscheduled (ADR-0009). [4, §4.2]: "Choosing B < A ensures that it is never favorable to violate the constraint", per weight here. [2, §4, Comment on the Scalar Penalty P]: P large enough that a hard constraint is never broken.
`anneal`	[5, cell 15]: simulated annealing (neal) on the BQM, then pyqubo decodes every read.
`qaoa`	[1, Eqs. (1)–(8)]: QAOA on the Ising form of the compiled model, simulated on Aer, with C = H and the expectation minimised, where Farhi maximises it.
`qaoa`	[4, §2.3, Eq. (12)]: pyqubo's spins, x = (s + 1)/2, so s = 2x - 1.
`qaoa`	[1, Eq. (5)]: |s>, the uniform superposition, top eigenstate of the mixer B
`qaoa`	[1, Eq. (6)]: p alternating cost and mixer layers.
`qaoa`	[1, Eq. (2)]: cost layer U(C, gamma) = exp(-i gamma C), C = H/scale. [4, §1.1, Eq. (4)]: on a qubit Z = 1 - 2x = -s, so a field h on s acts as -h on Z.
`qaoa`	[1, Eqs. (3)–(4)]: mixer layer U(B, beta) = exp(-i beta B), B = sum_j X_j.
`expected_energy`	[1, Eq. (7)]: F_p, the expectation of the cost in the state, estimated from the shots.
`qaoa`	[1, Eq. (8)]: the classical loop optimises the 2p angles.

Missing: The BESSER chain's rules and their references. Waits on the BESSER rework.
Once its code cites `[Key locator]` as the EMF chain's does, `generate.py` can read it too.

B Full results

Missing: These EMF numbers come from a local run, not from CI. Waits on CI.
Once CI has uploaded an `emf-results` artifact (push #24 and #25), rerun `build.sh`, or `build.sh --run <id>` for a pull request's run.

Table 3: Every run of the EMF chain: each test instance solved by both Methods, with the committed seeds; the Penalty Coefficient A used, the Schedule's cost and Unscheduled Exams, the Baseline's optimal cost, and P(optimal) and P(feasible) against a uniformly random bit string's. From a local run.

					Optimal	P(optimal)		P(feasible)	
Instance	Var.	Method	A	Result	cost	run	random	run	random
path	8	QAOA, depth 2, COBYLA, 150 iterations, 2048 shots	4	0, none	0	0.106	0.0078	0.533	0.160
path	8	annealing, 50 reads × 10000 sweeps	4	0, none	0	0.940	0.0078	1.000	0.160
cycle	15	QAOA, depth 2, COBYLA, 150 iterations, 4096 shots	2	0, none	0	0.033	0.00092	0.200	0.012
cycle	15	annealing, 50 reads × 10000 sweeps	2	0, none	0	1.000	0.00092	1.000	0.012
clique	12	annealing, 50 reads × 10000 sweeps	5	1, Statistics	1	0.940	0.0015	1.000	0.018
clique	12	QAOA, depth 2, COBYLA, 150 iterations, 4096 shots	5	1, Statistics	1	0.042	0.0015	0.553	0.018
semester	18	annealing, 50 reads × 10000 sweeps	4	0, none	0	0.960	0.000046	1.000	0.0034
semester	18	QAOA, depth 2, COBYLA, 150 iterations, 4096 shots	4	0, none	0	0.00024	0.000046	0.587	0.0034

Table 4: The Schedule each run of Table 3 found: the exams of each batch.

Instance	Method	Batch: exams	Unscheduled
path	QAOA	1: Calculus, Chemistry; 2: Algebra, Physics	none
path	annealing	1: Calculus, Chemistry; 2: Algebra, Physics	none
cycle	QAOA	1: Calculus; 2: Physics, Programming; 3: Algebra, Chemistry	none
cycle	annealing	1: Chemistry; 2: Calculus, Programming; 3: Algebra, Physics	none
clique	annealing	1: Physics; 2: Calculus; 3: Algebra	Statistics
clique	QAOA	1: Physics; 2: Calculus; 3: Algebra	Statistics
semester	annealing	1: Physics, Programming; 2: Calculus, Databases; 3: Algebra, Chemistry	none
semester	QAOA	1: Physics, Programming; 2: Calculus, Databases; 3: Algebra, Chemistry	none

Table 5: The EMF chain with other QAOA settings, every one reaching the optimal cost. Measured on macOS before `emf/requirements.txt` was pinned, so a rerun may differ in the second digit; copied by hand from `emf/CONTEXT.md`. COBYLA stops well before 150 iterations (27 to 71 evaluations); a Powell iteration is a line search over every angle (891 evaluations on cycle).

Instance	Variables	Method	P(optimal)	P(feasible)
path	8	QAOA, depth 1, COBYLA, 150 iterations, 2048 shots	0.012	0.403
path	8	QAOA, depth 3, COBYLA, 150 iterations, 2048 shots	0.009	0.500
path	8	QAOA, depth 2, Nelder-Mead, 150 iterations, 2048 shots	0.075	0.804
cycle	15	QAOA, depth 1, COBYLA, 150 iterations, 4096 shots	0.0007	0.006
cycle	15	QAOA, depth 3, COBYLA, 150 iterations, 4096 shots	0.039	0.451
cycle	15	QAOA, depth 2, Powell, 150 iterations, 4096 shots	0.132	0.512

Missing: The BESSER chain's full results. Waits on the BESSER rework.
Every instance's metrics, from `besser/output/summary.md` or its JSONs.

C Metamodels

Containment is drawn with a diamond at the container, a reference with an open arrow; *inv* names an OCL invariant.

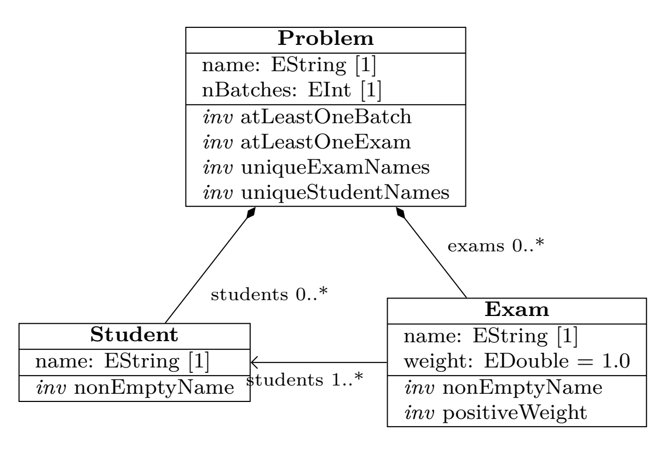


Figure 2: The problem metamodel, drawn from `problem.ecore`.

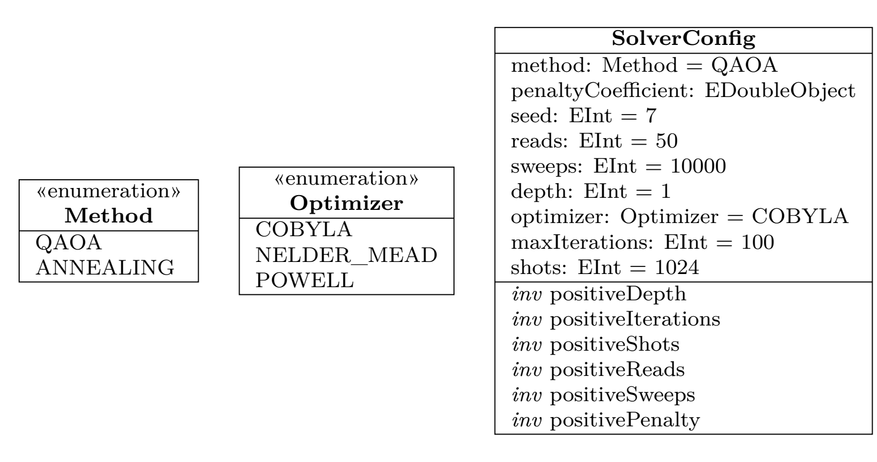


Figure 3: The solverconfig metamodel, drawn from `solverconfig.ecore`.

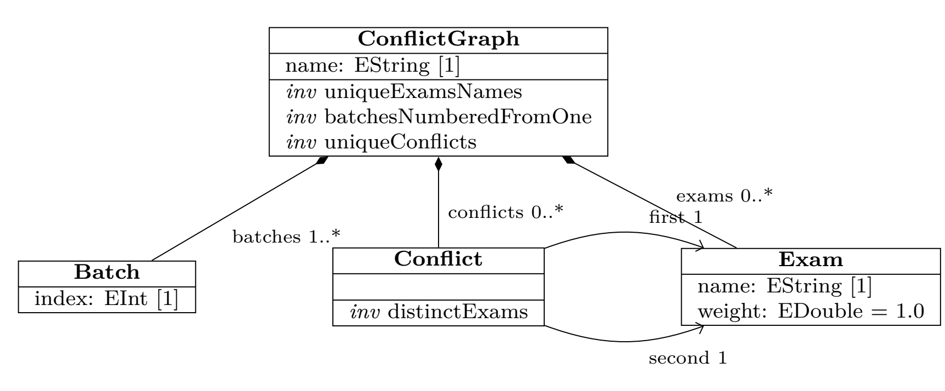


Figure 4: The conflictgraph metamodel, drawn from `conflictgraph.ecore`.

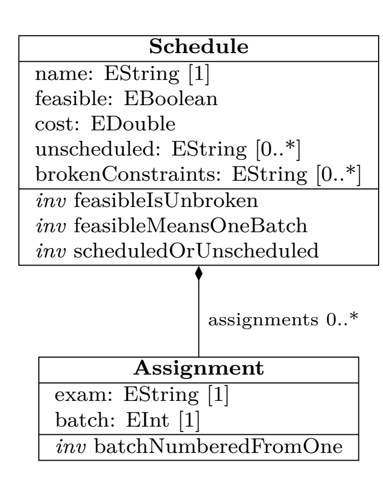


Figure 5: The schedule metamodel, drawn from `schedule.ecore`.

Missing: The BESSER chain's metamodels. Waits on the BESSER rework.
Part I's diagrams (`docs/imgs/problem-dsl-01.png`, `docs/imgs/solution-dsl-01.png`) once BESSER settles, with its Solver Configuration and quantum metamodels.

D Test instances

The EMF chain's instances, `emf/instances/`. Each has a twin, `<n>-<name>-qaoa.exam` or `<n>-<name>-annealing.exam`, holding the same problem solved by the other Method.

`1-path.exam`

```
// Q16 (i): four exams whose conflicts form a path, two batches. Feasible (a path is 2-colourable).
problem path batches 2
students Ana Rui Eva
exam Algebra   weight 2 { Ana }
exam Calculus  weight 3 { Ana Rui }
exam Physics   weight 1 { Rui Eva }
exam Chemistry weight 2 { Eva }

solver { method QAOA depth 2 optimizer COBYLA iterations 150 shots 2048 seed 7 }
```

`2-cycle.exam`

```
// Q16 (ii): five exams whose conflicts form a 5-cycle, three batches. Feasible (needs 3 batches).
// No weights: every exam weighs 1, so the schedule places as many exams as it can.
problem cycle batches 3
students Ana Rui Eva Ivo Lia
exam Algebra { Ana Lia }
exam Calculus { Ana Rui }
exam Physics { Rui Eva }
exam Chemistry { Eva Ivo }
exam Programming { Ivo Lia }

solver { method QAOA depth 2 optimizer COBYLA iterations 150 shots 4096 seed 7 }
```

`3-clique.exam`

```
// Q16 (iii): four exams that all conflict, three batches. No schedule places them all: the
// lowest-weight exam, Statistics, is left unscheduled, at cost 1 (ADR-0009).
problem clique batches 3
students Ana Rui Eva
exam Algebra    weight 4 { Ana Rui Eva }
exam Calculus   weight 3 { Ana Rui Eva }
exam Physics    weight 2 { Ana Rui Eva }
exam Statistics weight 1 { Ana Rui Eva }

solver { method annealing reads 50 sweeps 10000 seed 7 }
```

`4-semester.exam`

```
// Q16 (iv): six exams of a semester, three batches. Feasible.
problem semester batches 3
students Ana Rui Eva Ivo Lia Tom
exam Algebra     weight 3 { Ana Rui }
exam Calculus    weight 3 { Rui Eva Tom }
exam Physics     weight 2 { Ana Eva }
exam Chemistry   weight 2 { Ivo Lia }
exam Programming weight 1 { Lia Tom }
exam Databases   weight 1 { Ivo Ana }

solver { method annealing reads 50 sweeps 10000 seed 7 }
```

Missing: The BESSER chain's instances. Waits on the BESSER rework.
`besser/instances/`: tiny, many, none, medium and limit.

E Reproducing the results

- The EMF chain: `cd emf && ./build.sh` (JDK 21+, Python 3.12) builds the chain and runs it on every instance, writing `emf/out/<name>/results_<name>.json`. The Python packages, pyqubo [6], Qiskit and Qiskit Aer [3] and dwave-neal among them, are pinned in `emf/requirements.txt`. CI runs the same on every push (`.github/workflows/emf.yml`) and uploads the results as the `emf-results` artifact.
- This report: `docs/report_2/build.sh` (TeX Live, `gh`) fetches those results from CI, generates its tables, Section A and the diagrams of Section C, and runs latexmk; it does not run the chains.
- The LLM logs of Part II: `docs/llm_usage_2.md`, with the session records it links in `docs/assets/`, and `besser/docs/llm_usage_2.md`.

Missing: How to reproduce the BESSER chain's results. Waits on the BESSER rework.
`besser/README.md`'s commands, once the chain settles.

[1] BESSER's metamodels, from Part I, call a Batch a *TimeSlot* (or slot) and a feasible Schedule *valid*.